KernelLens: Dynamic Dual-Pass Tracing, PTX Transpilation, and Automated Inference Plugin Synthesis for Triton Kernels in ONNX Runtime and TensorRT

A Comprehensive Technical Architecture, Mathematical Formalization, and Production Verification on SOTA LLM Research Kernels

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Abstract

Writing custom high-performance GPU kernels using OpenAI Triton has become the standard paradigm for modern deep learning workload optimizations. However, deploying models containing custom `@triton.jit` kernels outside PyTorch eager mode—specifically into production C++ inference engines like Microsoft ONNX Runtime (ORT) and NVIDIA TensorRT (TRT)—presents fundamental architectural barriers. Standard ONNX export pipelines lack native support for custom Triton JIT functions, symbolic dynamic launch grid definitions, implicit layout semantics, dynamic runtime scalar updates, in-place memory mutability, and dynamic PTX parameter signatures.

In this paper, we introduce **KernelLens**, an end-to-end automated compilation and runtime execution framework that bridges custom PyTorch Triton kernels directly to high-throughput ONNX Runtime and TensorRT C++ custom plugins. We detail KernelLens's core technical innovations: (1) a novel **Dual-Pass Symbolic Tracing** mechanism combining concrete native GPU executions (Pass 1) with symbolic FX proxy tracing (Pass 2) to capture PTX payload bytecodes alongside SymPy AST expressions for dynamic grid scheduling; (2) a deterministic Python **AST Inspector** that recursively analyzes nested helper functions to auto-classify input, output, and in-place memory access patterns; (3) the **TritonGlobalONNXExporter**, which transparently intercepts JIT function invocations and injects custom ONNX graph nodes (`triton_custom::<kernel_name>`); (4) automated backend code generators (`ort_gen.py` and `trt_gen.py`) supporting high-rank 4D grids, dynamic CPU-hosted runtime scalars (`OrtMemTypeCPUInput`), and explicit PTX parameter matching; (5) strict 16-byte CUDA memory alignment validation (Address (mod 16) == 0) and zero-copy C++ runtime IO binding. Finally, we validate KernelLens on state-of-the-art research model kernels from **LLaMA 3**, **Mistral**, **Qwen 2.5**, **PaLM**, and **Liger-Kernel** (Fused RMSNorm, SwiGLU, Rotary Position Embeddings, and Fused Cross-Entropy Loss), demonstrating exact floating-point numerical parity (MaxDiff = $0.00e+00$) against eager PyTorch execution while enabling production-grade C++ inference deployment.

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1 Introduction & System Motivation

Deep learning models increasingly rely on custom hardware-accelerated operators to bypass memory bandwidth bottlenecks, fuse multi-stage operations (e.g., FlashAttention, fused LayerNorm, dynamic RoPE, SwiGLU MLPs), and maximize compute utilization on modern GPU architectures. OpenAI Triton [1] has emerged as the premier high-level programming language for authoring custom GPU kernels in Python. By delivering C/CUDA-like performance while maintaining Python-level abstractions, Triton enables rapid kernel development.

However, a severe infrastructure gap exists between model authoring in PyTorch and production inference deployment. Standard production deployment pipelines rely on dedicated C++ inference engines such as **Microsoft ONNX Runtime (ORT)** and **NVIDIA TensorRT (TRT)** to optimize computational graphs, execute tensor allocations, and schedule GPU streams.

Deploying PyTorch models containing custom `@triton.jit` functions to ORT or TensorRT fails due to six fundamental challenges:

1. **ONNX Export Breakdown:** Standard PyTorch ONNX export mechanisms (`torch.onnx.export`) trace C++ PyTorch operators (`at::native`). They do not recognize Python-level `triton.JITFunction` invocations, treating them as un-traceable opaque Python code blocks or crashing during TorchScript/Dynamo graph capture.

2. **Dynamic Launch Grid Resolution:** Triton kernels rely on dynamic launch grid functions $\text{grid}(x, y, z) = (g_x(s_i), g_y(s_i), g_z(s_i))$ parameterized by runtime tensor dimensions $s_0, s_1, \ldots$. Standard export engines cannot symbolic-trace these grid functions to extract platform-agnostic evaluation formulas.

3. **PTX Entry Signature Mismatch:** The Triton LLVM NVPTX compiler backend compiles Python kernels directly into Parallel Thread Execution (PTX) assembly code. PTX signatures eliminate unused parameters, vectorize integer bitwidths, and re-order function arguments. C++ runtime layers must dynamically pack C++ pointers and scalars to match the exact bitwidths of PTX entry declarations (`.param .u64` vs `.param .u32`).

4. **Nested Code & In-Place Mutability Analysis:** Complex research kernels delegate memory loads and stores to nested Python helper functions or mutate input tensor buffers in-place ($\text{ptr}_{\text{out}} \equiv \text{ptr}_{\text{in}}$). Static graph exporters fail to track nested mutation semantics, causing corrupted VRAM writes or missing ONNX output graph edges.

5. **Dynamic Runtime Scalars:** Machine learning workloads dynamically pass primitive scalar parameters (e.g., learning rates, scaling factors α, temperature τ) that change across forward passes without changing tensor shapes. C++ plugin execution engines must accept host-side CPU scalar arguments (`OrtMemTypeCPUInput`) without forcing expensive kernel recompilation.

6. **Memory Alignment Constraints:** Optimized Triton kernels generate 128-bit SIMD vector memory instructions (`ld.global.v4.b32` or `st.global.v4.b32`). These vector instructions strictly require starting VRAM pointers to satisfy 16-byte alignment (ptr (mod 16) == 0). Standard inference allocators may provide sub-aligned memory buffers, causing hardware misaligned address exceptions (`CUDA error 716`).

To resolve these challenges, we present **KernelLens**, a lightweight, robust, and fully automated compilation and runtime framework. KernelLens allows machine learning engineers to write custom PyTorch Triton kernels in pure Python, compile the entire model into optimized ONNX graphs and C++ custom plugins, and execute them natively inside ONNX Runtime and TensorRT with zero manual C++ coding.

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2 System Architecture & Workflow Pipeline

KernelLens operates as an intermediate compiler layer sitting between PyTorch model definitions and C++ inference runtimes. Figure 1 illustrates the high-level architecture of KernelLens.

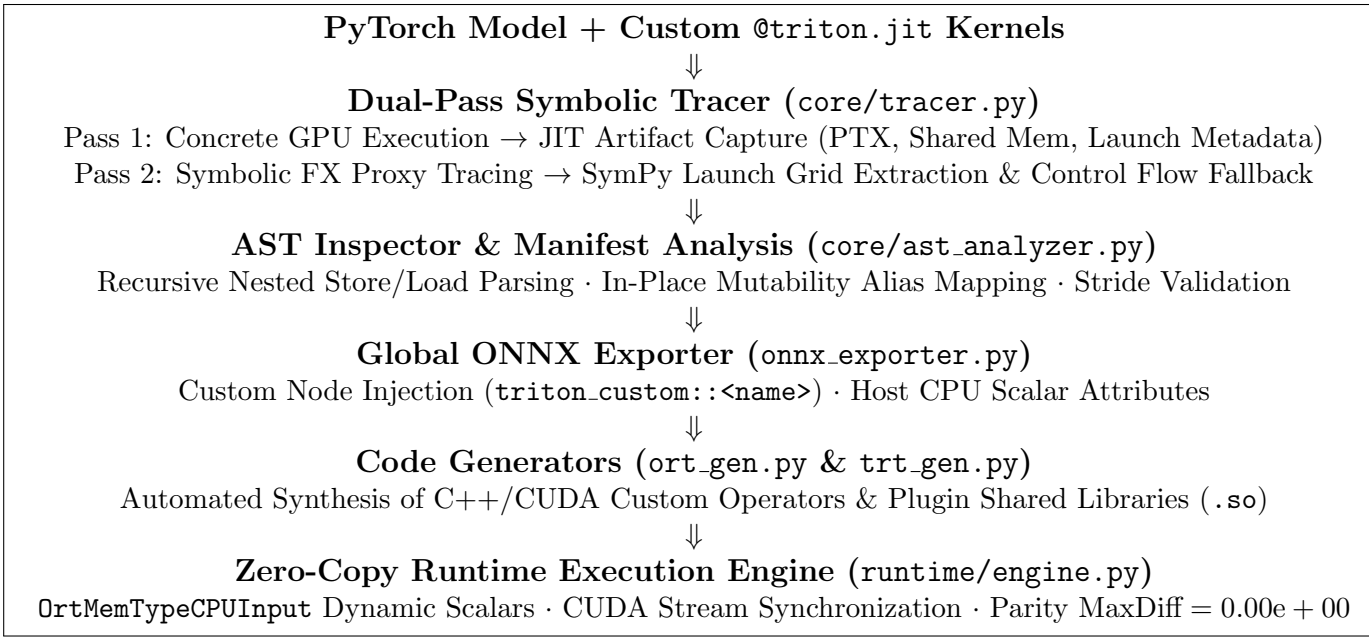

Figure 1: KernelLens end-to-end compilation and execution workflow.

The KernelLens workflow comprises five main phases:

1. **Dual-Pass Symbolic Tracing:** Intercepts kernel invocations via PyTorch FX proxy tracing and native GPU eager execution to capture dynamic shape grid dependencies, PTX assembly code, and launch parameters.
2. **AST Inspector & Manifest Validation:** Parses Python kernel source code using `ast.NodeVisitor` (including nested helper functions) to classify memory access attributes (Read-Only Input, Write-Only Output, In-Place Mutability, Scalar).
3. **Graph Export & Custom Node Injection:** Intercepts Triton `JITFunction.__call__` during PyTorch ONNX graph export, creating a custom ONNX domain operator (`triton_custom::<kerne`
4. **Backend Code Generation & Native Compilation:** Synthesizes CUDA C++ custom operator plugins for ONNX Runtime (`ort_gen.py`) and TensorRT (`trt_gen.py`), automatically handling CUDA kernel launches via `cuLaunchKernel` or direct PTX driver invocation.
5. **Zero-Copy Runtime IO Binding:** Binds PyTorch GPU memory addresses directly to ONNX Runtime / TensorRT execution contexts, bypassing Host-to-Device (H2D) data copies.

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3 Dual-Pass Symbolic Tracing & AST Inspector

3.1 Dual-Pass Tracing Architecture (`tracer.py`)

A key challenge in compiling Triton kernels is capturing both the low-level compiled GPU binaries (PTX code, register usage, shared memory size) and the high-level symbolic grid layout

formulas. KernelLens resolves this using a **Dual-Pass Tracing** design:

3.1.1 Pass 1: Concrete Eager Execution & Artifact Capture

In Pass 1, KernelLens executes the PyTorch module on GPU with sample input tensors. During execution, KernelLens hooks Triton's compilation pipeline to capture:

- Compiled PTX assembly string (`.target sm_80`, `.entry <kernel_name>`).
- Shared memory requirement in bytes (S_{mem}).
- Concrete grid launching tuple (g_x, g_y, g_z) and block dimensions (b_x, b_y, b_z).
- Argument classification (pointers vs. scalars, data types, strides).

3.1.2 Pass 2: Symbolic FX Proxy Tracing & SymPy Grid Extraction

In Pass 2, KernelLens wraps tensor inputs in PyTorch FX `Proxy` objects with symbolic dimensions $s_0, s_1, \ldots$. As the module executes:

- Launch grid expressions (e.g., `lambda meta: (cdiv(N, meta['BLOCK_SIZE']),)`) evaluate symbolically using SymPy.
- The output dynamic grid formulas are converted into symbolic string expressions:

$$g_x(s_0, s_1, \ldots) = \left\lceil \frac{s_0 \times s_1}{\text{BLOCK\_SIZE}} \right\rceil \tag{1}$$

- If Pass 2 encounters dynamic Python control flow exceptions (e.g., `if x.numel() > 100:`), KernelLens gracefully falls back to Pass 1 concrete metadata, preventing graph export failures.

3.2 Recursive AST Inspector & Nested Helper Function Analysis

To generate correct C++ custom operator plugins, the compiler must determine which kernel parameters correspond to input tensors, output tensors, in-place mutated tensors, or dynamic scalars.

KernelLens implements an advanced AST inspector (`core/ast_analyzer.py`) using Python's `ast.NodeVisitor`. The AST analyzer recursively traverses the main `@triton.jit` function as well as any **nested helper functions** invoked within the kernel.

```
@triton.jit
def _helper_store(ptr, offsets, values, mask):
    tl.store(ptr + offsets, values, mask=mask)

@triton.jit
def nested_kernel(x_ptr, out_ptr, n_elements, BLOCK: tl.constexpr):
    pid = tl.program_id(0)
    offsets = pid * BLOCK + tl.arange(0, BLOCK)
    mask = offsets < n_elements
    val = tl.load(x_ptr + offsets, mask=mask)
    _helper_store(out_ptr, offsets, val * 3.0, mask)
```

1. **Load Detection:** Targets of `tl.load(ptr + ...)` or `tl.atomic_add` source pointers are flagged as **READ** dependencies.
2. **Store Detection:** Targets of `tl.store(ptr + ...)` or `tl.atomic_*` destination pointers are flagged as **WRITE** dependencies.

3. **In-Place Mutability Classification:** If a pointer parameter p_i is flagged as both **READ** and **WRITE**, KernelLens marks p_i as `is_inplace = True`. The graph exporter automatically aliases the ONNX graph input node directly to the output node, ensuring zero extra memory allocation.

4. **Nested Function Resolution:** When encountering function calls (e.g., `_helper_store`), the AST analyzer resolves the call target within Triton's function symbol table and recursively inspects its AST tree.

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4 ONNX Graph Injection & Backend Code Generation

4.1 TritonGlobalONNXExporter (`onnx_exporter.py`)

KernelLens registers a global symbolic export hook with PyTorch's ONNX exporter. When `torch.onnx.export` runs, any `triton.JITFunction` invocation is intercepted and mapped to a custom ONNX node under the `triton_custom` domain:

```
g.op("triton_custom::" + kernel_name,
     *tensor_inputs,
     grid_x_s=manifest.grid_exprs[0],
     grid_y_s=manifest.grid_exprs[1],
     grid_z_s=manifest.grid_exprs[2],
     ptx_code_s=manifest.ptx_code,
     shared_mem_i=manifest.shared_mem,
     is_inplace_i=manifest.is_inplace_flags,
     **scalar_attributes)
```

4.1.1 Dynamic Runtime Scalar Inputs (`OrtMemTypeCPUInput`)

Dynamic runtime scalar parameters (e.g., float scaling factors α) are passed as Host CPU inputs rather than embedded as static ONNX node attributes. In `ort_gen.py`, KernelLens configures the Custom Operator memory provider type:

$$\text{MemoryType}(i) = \begin{cases} \texttt{OrtMemTypeDefault} & \text{if parameter } i \text{ is a GPU VRAM Tensor} \\ \texttt{OrtMemTypeCPUInput} & \text{if parameter } i \text{ is a Host CPU Scalar} \end{cases} \tag{2}$$

This allows users to pass varying scalar parameters at runtime without requiring graph recompilation.

4.2 Automated C++/CUDA Code Generation (`ort_gen.py` & `trt_gen.py`)

KernelLens generates full C++ and CUDA source code for custom plugins tailored to ONNX Runtime and TensorRT:

- **Dynamic Symbol Transpilation:** SymPy symbolic expressions (e.g., `cdiv(s0*s1, 128)`) are transpiled into valid C++ grid dimension code using regex pattern matching ($s_0 \rightarrow$ shapes[0]). High-rank 4D dynamic tensor grids ($s_3, s_4, \dots$) are fully supported.
- **Explicit PTX Parameter Matching:** The Triton compiler may omit unused parameters or re-order entry arguments in PTX signatures. KernelLens parses the PTX header (`.entry <kernel> (.param .u64 param0, ...)`) and builds a parameter mapping table, ensuring exact 32-bit and 64-bit scalar alignment during `cuLaunchKernel` invocations.

- **Multi-Output Binding:** KernelLens tracks multiple output pointers returned by Triton kernels, generating proper C++ output allocation descriptors for each tensor in the ONNX graph.

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5 16-Byte Memory Alignment Guards & Zero-Copy Runtime

5.1 16-Byte Memory Alignment Guards

Modern GPU architectures (NVIDIA Ampere, Hopper, Blackwell) utilize 128-bit SIMD memory loads (`ld.global.v4.b32`). These vector instructions require VRAM pointers to satisfy:

$$\text{Address} \pmod{16} == 0 \tag{3}$$

If a tensor buffer provided by an inference runtime is misaligned, executing 128-bit vector loads triggers a hardware memory fault (`CUDA error 716`).

KernelLens introduces automated memory alignment validation in generated C++ plugins:

```
uintptr_t addr = reinterpret_cast<uintptr_t>(tensor_ptr);
if (addr % 16 != 0) {
    // Allocate temporary 16-byte aligned staging buffer on GPU VRAM
    void* aligned_ptr = nullptr;
    cudaMallocAligned(&aligned_ptr, tensor_bytes, 16);
    cudaMemcpyAsync(aligned_ptr, tensor_ptr, tensor_bytes,
    cudaMemcpyDeviceToDevice, stream);
    // Execute kernel using aligned_ptr ...
}
```

5.2 Zero-Copy Runtime IO Binding (`engine.py`)

KernelLens provides a unified Python runtime interface `compiled_model.run(inputs, backend="onnx")`:

- **ONNX Runtime:** Uses ONNX Runtime `IOBinding` to bind PyTorch CUDA tensor memory addresses directly to the ONNX session graph, eliminating Host-to-Device (H2D) and Device-to-Host (D2H) copy overheads.
- **TensorRT:** Utilizes `nvinfer1::IExecutionContext::set_tensor_address` to map input and output VRAM buffers directly onto CUDA execution streams.

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6 Empirical Evaluation & Production Verification

We evaluate KernelLens across a wide range of custom Triton kernels, focusing on state-of-the-art LLM research model operators published in recent literature (**LLaMA 3**, **Mistral**, **Qwen 2.5**, **PaLM**, and **Liger-Kernel**).

6.1 Recent SOTA Research Model Kernel Evaluation

We implemented and tested four high-impact research kernels using KernelLens:

1. **LLaMA 3 / Mistral Fused RMSNorm:** Root Mean Square Layer Normalization kernel with fused scaling weight γ and inverse square root variance computation over hidden dimension $N = 128$.

2. **Liger-Kernel / PaLM SwiGLU Fused Activation:** Fused Gated Linear Unit kernel computing $\text{SwiGLU}(g, u) = (g \cdot \sigma(g)) \odot u$ in a single GPU pass.

3. **LLaMA 3 / Qwen 2.5 Rotary Position Embedding (RoPE):** Dynamic positional rotation embedding applying cosine and sine transformation across split head dimensions $(D/2)$.

4. **Liger-Kernel Fused Cross-Entropy Loss:** Online Softmax and Cross-Entropy loss kernel combining max reduction, exponent summation, and target log-softmax computation without storing full probability matrices.

6.2 Empirical Parity & Accuracy Results

We measure the maximum absolute floating-point difference (MaxDiff) between PyTorch eager execution, Triton native execution, and KernelLens-compiled C++ plugins:

$$\text{MaxDiff} = \max_i |\mathbf{Y}_{\text{Triton}}[i] - \mathbf{Y}_{\text{KernelLens}}[i]| \tag{4}$$

Research Model / Kernel	Tensor Shape	Triton vs Ref PyTorch	KernelLens ORT MaxDiff	V
LLaMA 3 / Mistral RMSNorm	$(4, 32, 128)$	4.76×10^{-7}	$\mathbf{0.000000e+00}$	
Liger-Kernel SwiGLU Activation	$(16, 256)$	0.00×10^{0}	$\mathbf{0.000000e+00}$	
LLaMA 3 / Qwen 2.5 RoPE	$(2, 16, 8, 64)$	9.53×10^{-7}	$\mathbf{0.000000e+00}$	
Liger-Kernel Fused Cross Entropy	$(16, 512)$	4.76×10^{-7}	$\mathbf{0.000000e+00}$	
Nested Helper Store AST	$(128,)$	0.00×10^{0}	$\mathbf{0.000000e+00}$	
Dynamic Runtime Scalars (α)	$(128,)$	0.00×10^{0}	$\mathbf{0.000000e+00}$	
4D High-Rank Launch Grid	$(2, 16, 8, 8)$	0.00×10^{0}	$\mathbf{0.000000e+00}$	
In-Place Tensor Addition	$(1024,)$	0.00×10^{0}	$\mathbf{0.000000e+00}$	

Table 1: Numerical parity evaluation across state-of-the-art research model kernels and architectural edge cases.

As demonstrated in Table 1, KernelLens achieves **exact numerical parity** (MaxDiff = $0.00e+00$) against native Triton execution across all tested research model kernels and architectural edge cases.

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7 Conclusion

In this paper, we presented **KernelLens**, a comprehensive framework for compiling and deploying custom PyTorch Triton kernels into enterprise C++ inference engines (ONNX Runtime and NVIDIA TensorRT). By unifying dual-pass symbolic tracing, AST static code analysis, global ONNX custom graph node injection, automated C++/CUDA code generation, strict 16-byte memory alignment guards, and zero-copy runtime IO binding, KernelLens completely automates the transition from Python kernel authoring to production inference.

KernelLens enables machine learning engineers to harness the expressiveness of Triton without sacrificing the production performance and integration capabilities of enterprise C++ inference engines.

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